

Homework 5

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5-2. 推迟的场叶菲缅科 (Jefimenko) 公式. (1)

Proof. 真空中的麦克斯韦方程

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0, \quad (1.1)$$

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}, \quad (1.2)$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \quad (1.3)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (1.4)$$

对 (1.3) 式取旋度得到

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} + \frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = 0,$$

再代入 (1.1) 式和 (1.2) 式,

$$\begin{aligned} & \frac{1}{\epsilon_0} \nabla \rho - \nabla^2 \mathbf{E} + \frac{\partial}{\partial t} \left[\frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right] = 0 \\ \Rightarrow & \left[\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \mathbf{E}(\mathbf{x}, t) = -\frac{1}{\epsilon_0} \left[-\nabla \rho - \frac{1}{c^2} \frac{\partial \mathbf{J}}{\partial t} \right]. \end{aligned} \quad (1.5)$$

同样地, 对 (1.2) 式取旋度并结合 (1.4) 得到

$$-\nabla^2 \mathbf{B} - \frac{1}{c^2} \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = \mu_0 \nabla \times \mathbf{J},$$

代入 (1.3) 式,

$$\left[\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \mathbf{B}(\mathbf{x}, t) = -\mu_0 \nabla \times \mathbf{J}. \quad (1.6)$$

□

(2) 利用 Fourier 变换可以解出符合物理因果律的推迟格林函数

$$G^{(+)}(R, \tau) = \frac{\delta(t' - [t - \frac{R}{c}])}{R},$$

由此可以积分得到波动方程的解为

$$\mathbf{E}(\mathbf{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3\mathbf{x}' dt' \frac{\delta(t' - [t - \frac{R}{c}])}{R} \left[-\nabla' \rho - \frac{1}{c^2} \frac{\partial \mathbf{J}}{\partial t'} \right] = \frac{1}{4\pi\epsilon_0} \int d^3\mathbf{x}' \frac{1}{R} \left[-\nabla \rho - \frac{1}{c^2} \frac{\partial \mathbf{J}}{\partial t} \right]_{\text{ret}}, \quad (1.7)$$

$$\mathbf{B}(\mathbf{x}, t) = \frac{\mu_0}{4\pi} \int d^3\mathbf{x}' dt' \frac{\delta(t' - [t - \frac{R}{c}])}{R} [\nabla' \times \mathbf{J}] = \frac{\mu_0}{4\pi} \int d^3\mathbf{x}' \frac{1}{R} [\nabla' \times \mathbf{J}]_{\text{ret}}. \quad (1.8)$$

4. 推导长波近似下的电偶极辐射电磁场。书 (5.19) 式给出了长波近似下的磁矢势

$$\mathbf{A}(\mathbf{x}) = -\frac{i\mu_0\omega}{4\pi} \mathbf{p} \frac{e^{ikr}}{r}. \quad (5.19)$$

利用书 (5.14) 式以及书 (5.20) 式给出的矢量分析公式，有磁场

$$\begin{aligned} \mathbf{H} &= \frac{1}{\mu_0} \nabla \times \mathbf{A} = -\frac{i\omega}{4\pi} \nabla \times \left(\mathbf{p} \frac{e^{ikr}}{r} \right) = -\frac{i\omega}{4\pi} \nabla \frac{e^{ikr}}{r} \times \mathbf{p} \\ &= \frac{ck^2}{4\pi} (\mathbf{n} \times \mathbf{p}) \frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr} \right). \end{aligned} \quad (1.9)$$

进而有

$$\begin{aligned} \mathbf{E} &= \frac{iZ_0}{k} \nabla \times \mathbf{H} = \frac{ik}{4\pi\epsilon_0} \nabla \times \left[(\mathbf{n} \times \mathbf{p}) \frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr} \right) \right] \\ &= \frac{ik}{4\pi\epsilon_0} \left\{ \nabla \left[\frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr} \right) \right] \times (\mathbf{n} \times \mathbf{p}) + \frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr} \right) \nabla \times (\mathbf{n} \times \mathbf{p}) \right\} \\ &= \frac{ik}{4\pi\epsilon_0} \left\{ \left(ik - \frac{2}{r} + \frac{2}{ikr^2} \right) \frac{e^{ikr}}{r} [\mathbf{n} \times (\mathbf{n} \times \mathbf{p})] + \frac{1}{ik} \left(\frac{1}{r^3} - \frac{ik}{r^2} \right) [\mathbf{n}(\mathbf{n} \cdot \mathbf{p}) - \mathbf{p}] \right\} \\ &= \frac{1}{4\pi\epsilon_0} \left\{ k^2 (\mathbf{n} \times \mathbf{p}) \times \mathbf{n} \frac{e^{ikr}}{r} + [3\mathbf{n}(\mathbf{n} \cdot \mathbf{p}) - \mathbf{p}] \left(\frac{1}{r^3} - \frac{ik}{r^2} \right) e^{ikr} \right\}. \end{aligned} \quad (1.10)$$

郭书 5.10 在直角坐标中可以将磁偶极矩表示为

$$\mathbf{m} = \frac{4\pi}{3} M_0 R_0^3 e^{-i\omega t} (\mathbf{e}_x + i\mathbf{e}_y),$$

转换到球坐标中，

$$\mathbf{m} = \frac{4\pi}{3} M_0 R_0^3 e^{-i\omega t + i\phi} (\sin \theta \mathbf{e}_R + \cos \theta \mathbf{e}_\theta + i\mathbf{e}_\phi).$$

带入辐射电磁场磁偶极项表达式

$$\begin{aligned}\mathbf{B} &= \frac{\mu_0 \mathbf{e}^{ikR}}{4\pi c^2 R} (\ddot{\mathbf{m}} \times \mathbf{e}_R) \times \mathbf{e}_R = \frac{\mu_0 \omega^2 R_0^3 M_0}{3c^2 R} (\mathbf{e}_\theta \cos \theta + i\mathbf{e}_\phi) e^{i(kR - \omega t + \phi)}, \\ \mathbf{E} &= -\frac{\mu_0 \mathbf{e}^{ikR}}{4\pi c R} (\ddot{\mathbf{m}} \times \mathbf{e}_R) = \frac{\mu_0 \omega^2 R_0^3 M_0}{3c R} (i\mathbf{e}_\theta - \mathbf{e}_\phi \cos \theta) e^{i(kR - \omega t + \phi)},\end{aligned}$$

故有

$$\mathbf{S} = \frac{1}{2\mu_0} \text{Re}(\mathbf{E}^* \times \mathbf{B}) = \frac{\mu_0 \omega^4 R_0^6 M_0^2}{18c^3 R^2} (1 + \cos^2 \theta) \mathbf{e}_R.$$

郭书 5.11 直角坐标中电偶极矩表示为

$$\mathbf{p} = ea e^{-i\omega t} (\mathbf{e}_x + i\mathbf{e}_y),$$

相应的球坐标表示为

$$\mathbf{p} = ea e^{-i\omega t + i\phi} (\sin \theta \mathbf{e}_R + \cos \theta \mathbf{e}_\theta + i\mathbf{e}_\phi).$$

带入辐射电磁场电偶极项表达式

$$\begin{aligned}\mathbf{E} &= \frac{\mathbf{e}^{ikR}}{4\pi \epsilon_0 c^2 R} (\ddot{\mathbf{p}} \times \mathbf{e}_R) \times \mathbf{e}_R = \frac{\omega^2 ea}{4\pi \epsilon_0 c^2 R} (\mathbf{e}_\theta \cos \theta + i\mathbf{e}_\phi) e^{i(kR - \omega t + \phi)}, \\ \mathbf{B} &= \frac{\mathbf{e}^{ikR}}{4\pi \epsilon_0 c^3 R} \ddot{\mathbf{p}} \times \mathbf{e}_R = \frac{\omega^2 ea}{4\pi \epsilon_0 c^3 R} (\mathbf{e}_\phi \cos \theta - i\mathbf{e}_\theta) e^{i(kR - \omega t + \phi)},\end{aligned}$$

辐射能流

$$\mathbf{S} = \frac{1}{2\mu_0} \text{Re}(\mathbf{E}^* \times \mathbf{B}) = \frac{\mu_0 \omega^4 e^2 a^2}{32\pi^2 c R^2} (1 + \cos^2 \theta) \mathbf{e}_R.$$

8-1. 运动的电偶极子.

(1) 直接计算系统的电偶极矩和磁偶极矩得到

$$\begin{aligned}p'_i &= \int d^3 \mathbf{x}' x'_i \rho'(\mathbf{x}') = - \int d^3 \mathbf{x}' x'_i (\nabla' \cdot \mathbf{p}) \delta^3(\mathbf{x}') \\ &= \int d^3 \mathbf{x}' x'_i [\delta^3(\mathbf{x}') \nabla' \cdot \mathbf{p} - \nabla' \cdot (\mathbf{p} \delta^3(\mathbf{x}'))] \\ &= -x'_i \mathbf{p} \delta^3(\mathbf{x}') \Big|_{\Sigma} + \int d^3 \mathbf{x}' (\nabla' x'_i) \cdot \mathbf{p} \delta^3(\mathbf{x}') \\ &= \int d^3 \mathbf{x}' p_i \delta^3(\mathbf{x}') = p_i, \\ \mathbf{m}' &= \frac{1}{2} \int d^3 \mathbf{x}' [\mathbf{x}' \times \mathbf{J}'(\mathbf{x}')] = 0,\end{aligned}$$

即 K' 系中的电偶极矩为 $\mathbf{p}$, 磁偶极矩为 0.

(2) K' 系中电偶极矩四矢量可以表示为 $P'^\mu = (P'^0, \mathbf{P}') = (0, \mathbf{p})$, K 系相对 K' 系以 $-\boldsymbol{\beta}$ 运动, 应用书 (6.8) 式可以得到

$$P^0 = \gamma(P'^0 + \boldsymbol{\beta} \cdot \mathbf{P}') = \gamma\boldsymbol{\beta} \cdot \mathbf{p}, \quad \mathbf{P} = \mathbf{P}' + \frac{\gamma-1}{\beta^2}(\boldsymbol{\beta} \cdot \mathbf{P}')\boldsymbol{\beta} + \gamma\boldsymbol{\beta}P'^0 = \mathbf{p} + \frac{\gamma-1}{\beta^2}(\boldsymbol{\beta} \cdot \mathbf{p})\boldsymbol{\beta}.$$

故 K 系中电偶极矩四矢量为 $P^\mu = (\gamma\boldsymbol{\beta} \cdot \mathbf{p}, \mathbf{p} + \frac{\gamma-1}{\beta^2}(\boldsymbol{\beta} \cdot \mathbf{p})\boldsymbol{\beta})$. 另外, K 系中四速度为 $u_\mu = \gamma(1, -\boldsymbol{\beta})$, 二者内积为

$$P \cdot u = P^\mu u_\mu = \gamma[\gamma\boldsymbol{\beta} \cdot \mathbf{p} - \boldsymbol{\beta} \cdot \mathbf{p} - (\gamma-1)\boldsymbol{\beta} \cdot \mathbf{p}] = 0,$$

即在任意 K 系中 P^μ 均与四速度正交.

(3) 在静止系 K' 中, 积分表达式可写为

$$J^\mu(x) = - \int d\tau (1, 0)(\mathbf{p} \cdot \nabla')\delta(t' - \tau)\delta^3(\mathbf{x}') = -(\mathbf{p} \cdot \nabla')\delta^3(\mathbf{x}'), 0$$

对比各项表达式即得到该协变表达式的确给出 (1) 中结果.

(4) 由书 (8.8) 式, 代入 $J^\mu(x')$ 的协变表达式得到

$$A^\mu(x) = \int d^4x' D^{(+)}(x-x')J^\mu(x') = -4\pi \int d\tau u^\mu(\tau) \int d^4x' D^{(+)}(x-x')(P \cdot \partial')\delta^4[x'-r(\tau)].$$

关于 x' 的项分部积分, 边界项消失

$$\int d^4x' D^{(+)}(x-x')\partial'_\nu\delta^4[x'-r(\tau)] = - \int d^4x' \delta^4[x'-r(\tau)]\partial'_\nu D^{(+)}(x-x'),$$

注意到 $\partial'_\nu D^{(+)}(x-x') = -\partial_\nu D^{(+)}(x-x')$, 因此我们得到

$$\int d^4x' D^{(+)}(x-x')(P \cdot \partial')\delta^4[x'-r(\tau)] = (P \cdot \partial) \int d^4x' \delta^4[x'-r(\tau)]D^{+}(x-x') = (P \cdot \partial)D^{(+)}[x-r(\tau)].$$

带回 A^μ 表达式

$$A^\mu(x) = -4\pi \int d\tau u^\mu(\tau)(P \cdot \partial)D^{(+)}[x-r(\tau)].$$

进而我们可以代入 $D^{(+)}[x-r(\tau)]$ 的具体形式

$$A^\mu(x) = -2 \int d^4\tau u^\mu(\tau)\theta(x_0 - x'_0)(P \cdot \partial)\delta[(x-r(\tau))^2].$$

应用题中所给公式

$$\partial^\nu \delta[(x-r(\tau))^2] = \partial^\nu (x-r(\tau))^2 \frac{d\tau}{d(x-r(\tau))^2} \frac{d}{d\tau} \delta[(x-r(\tau))^2] = -\frac{(x-r(\tau))}{(x-r(\tau)) \cdot u} \frac{d}{d\tau} \delta[(x-r(\tau))^2],$$

带回前一个表达式得到

$$A^\mu(x) = 2 \int d^4\tau u^\mu(\tau) \theta(x_0 - x'_0) \frac{(x - r(\tau)) \cdot P}{(x - r(\tau)) \cdot u} \frac{d}{d\tau} \delta[(x - r(\tau))^2],$$

再作分部积分并运用 **delta** 函数性质得到最终表达式

$$A^\mu(x) = - \int d\tau \delta[(x - r(\tau))^2] \theta(x_0 - r_0) \frac{d}{d\tau} \left[\frac{u^\mu \cdot (P \cdot (x - r))}{u \cdot (x - r)} \right] = - \frac{1}{u \cdot R} \frac{d}{d\tau} \left[\frac{u^\mu (P \cdot R)}{u \cdot R} \right]_{\text{ret}},$$

其中 $R^\mu = x^\mu - r^\mu(\tau)$ 为场点到源点的相对坐标四矢量.